Technical Skill Debrief: Parametric Enclosure Design

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1. Introduction

This document outlines the workflow I developed for designing a custom battery enclosure for the McMaster Rocketry payload bay using Onshape. The process required implementing version control, master model methodology, wiring and space allocation planning, and design-for-manufacturing considerations specific to 3D printing. This guide summarizes the modeling tools used, key constraints and decisions, and the reasoning behind design iterations (Versions 1-3). It is written both as both a technical guide to the tools and methods I used, and a record of the design evolution for future team members working on similar pieces.

2. How-To Guide: Designing a Parametric Electronics Enclosure in Onshape

2.1 Set Up the CAD Workspace and Versioning

A successful enclosure design begins with an organized Onshape document. To ensure clarity and easy collaboration, I followed the team's preferred structure of separating the project into documents containing part studios, assemblies, and features. Hardware imported from external sources, such as components off the shelf, were kept in separate documents to simplify derivations and improve performance. For projects specific to McMaster Rocketry, make sure that everything abides to the respective space allocation (SA) volumes.

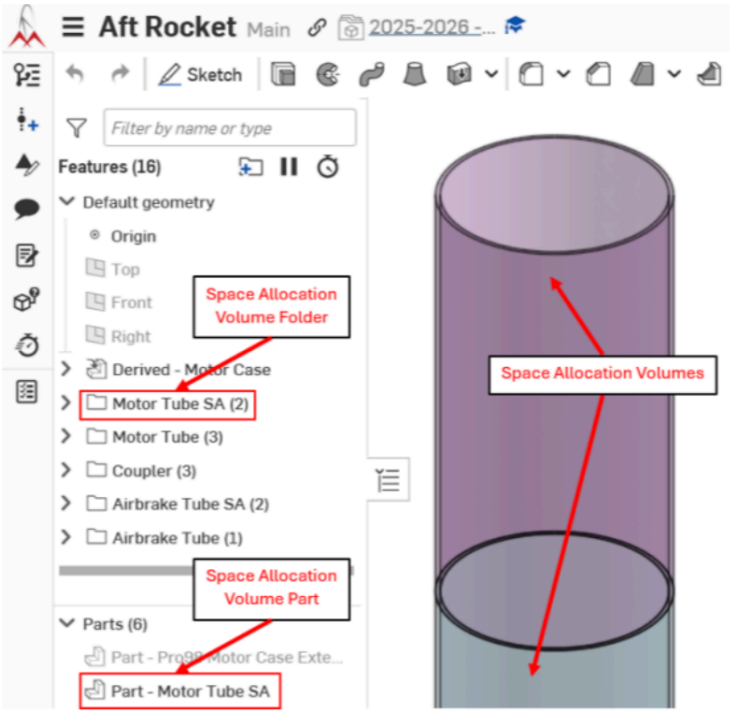


Figure 1: Example SA for the Motor Tube.

Version control quickly became one of the most valuable practices in the workflow. After each major design change, whether it involved the enclosure geometry, wiring allocation, or derived references, I created a new version. Referencing versions instead of the active workspace prevents unexpected downstream updates and allows intentional propagation of changes. This approach is especially crucial when using derived sketches or parts, because it ensures that updates to the master model only affect the design when explicitly approved.

To make the document maintainable, I renamed all sketches, extrusions, mate connectors, and parts following a clear naming format. Systematic labelling made editing, revisiting decisions, and collaborating with teammates significantly easier.

1. ***Use the structured document hierarchy***
   Organize your project according to the team's established structure:
   Project → Folders → Documents → Assemblies Part Studios → Part → Part Features.

   Keep payload hardware, imported objects, and derived references in separate documents for performance and clarity.

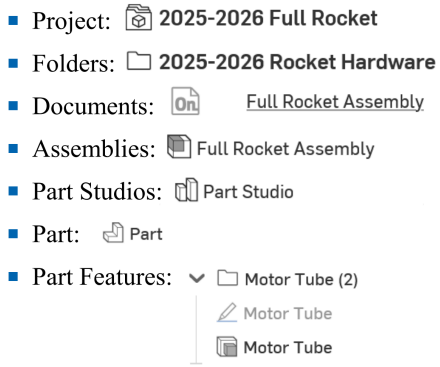


Figure 2: Example document hierarchy for the Motor Tube feature

2. ***Create versions early and often.***
   After major updates (e.g., wiring layout, derived parts, hole placement), create a version to lock a snapshot. Reference versions, not workspaces, when deriving parts so changes propagate only when intended. Be sure to avoid chains of derived instances - always derive from the original master model.

3. ***Naming conventions.***
   Rename every feature and part using Shift + N, following the format:
   "Sketch / Extrude / Mate Connector / Part" - "part's name".

   For example, a sketch of the Motor Tube would be: "Sketch - Motor Tube". The respective folder could simply hold all the features and be named: "Motor Tube".

2.2 Master Model Workflow

The master model served as the backbone of the design. In a dedicated part studio, I created master sketches to define driving dimensions, battery clearances, PCB footprints, and the wiring envelope. These sketches captured the critical relationships and constraints before any 3D geometry was created. By defining variables for frequently modified dimensions, such as wall thickness, shadow line spacing, and enclosure height, I could adjust the enclosure rapidly without manually editing each feature.

Once the master sketches were finalized, I used the Derive tool to bring them into downstream part studios. This allowed me to build detailed geometry while maintaining a stable reference to the original design intent. Importantly, I derived from a fixed version rather than the workspace, ensuring that later changes to the master model did not unintentionally break the enclosure or mounting features. This workflow kept the design modular, controlled, and updateable without inducing cascading errors.

Use a Master Sketch Part Studio to define:

I. Driving parameters (critical dimensions, enclosure height, minimum wall thickness).
II. Space allocations (battery volume, PCB stack footprint, wire routing envelope).
III. Layout sketches for mate connector locations and assembly references.

Derived Workflow

I. Derive the master sketch or key parts into child part studios.
II. Edits made in the master will update anywhere that references the workspace, but not version references.
III. Recommended: derive from the original instance to stabilize downstream models. Do NOT create a chain of derived instances.

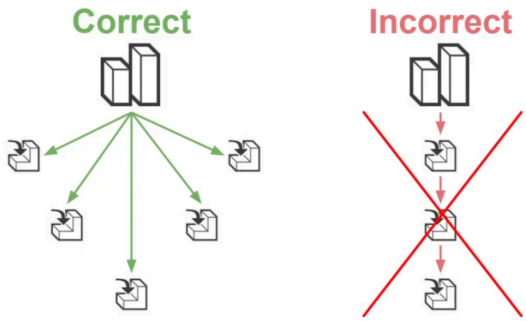


Figure 3: Correct use of Derived Workflow.

2.3 Design Considerations for Electronics Enclosures

Throughout the design process, several practical considerations shaped the enclosure. Because the internal components include wires, connectors, four 14500 batteries, and a PCB stack, I allocated explicit space for each. Wire routing received significant attention: I left 15 mm from the wire exit hole to the wall of the payload bay to allocate space for the wire connector, and ensured enough room for 1.9 mm diameter

wires. Small cutouts at the rear of the enclosure provided a wire routing path, and a hole for the wire connector was added in the second design revision.

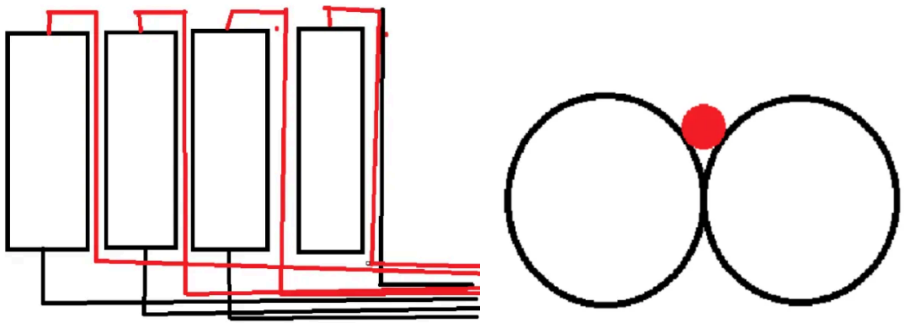


Figure 4: Concept sketches for wiring routing.

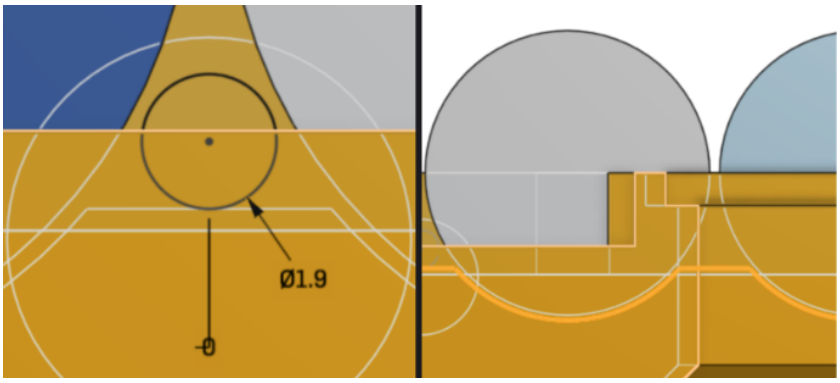


Figure 5: Wiring allowance incorporated in the final design.

To ensure the printed enclosure was structurally robust, it followed a minimum wall thickness of 1.6 mm. In Version 2 to improve print quality and eliminate sagging on horizontal surfaces, I replaced fillets with chamfers.

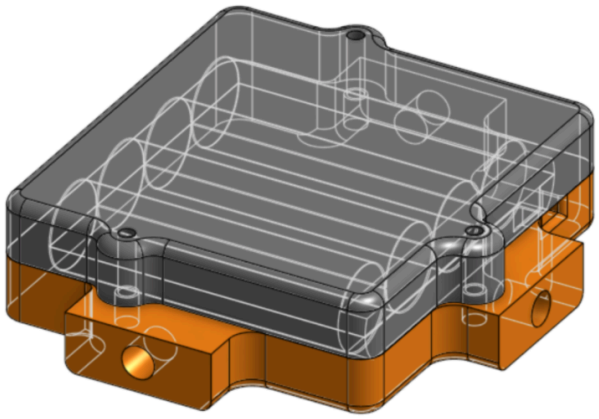


Figure 6: Version 1 of the Battery Case incorporated fillets.

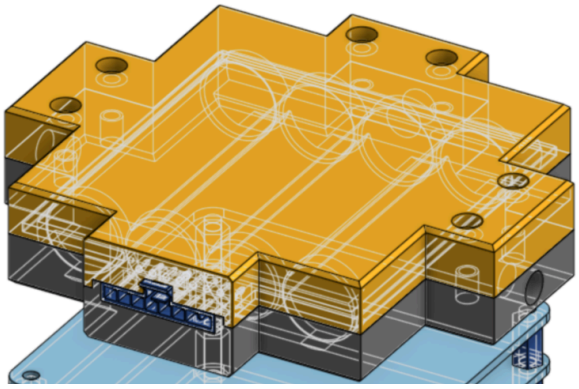


Figure 7: Version 2 used chamfers instead of fillets, and incorporated a wire connector.

Shadow lines were also incorporated with a standard 1.6 mm gap, based on best practices for 3D printed parts. This feature not only improved the appearance of the print but also strengthened the edges by redistributing material.

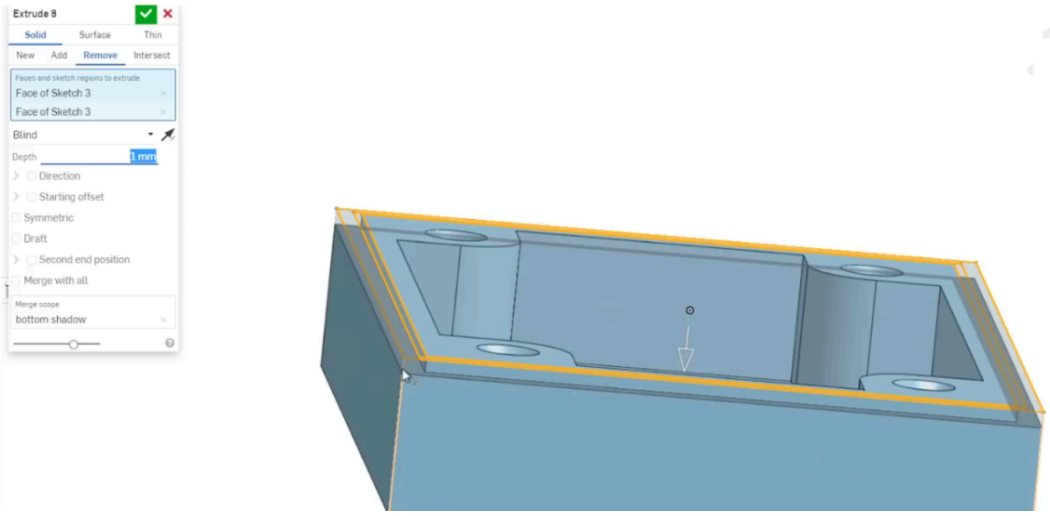


Figure 8: Example of extruding shadow lines.

The fastening strategy was another key constraint. I incorporated holes for heat inserts and added counterbores matched to screw head diameters. Note that washers are not required when the counterbore feature is implemented. To simplify assembly, I minimized the number of fasteners that needed to anchor to payload bay struts and focused on designing an enclosure that offered easy access to batteries without requiring the PCB stack to be disassembled. This also required ensuring enough material existed around heat-insert locations and reserving space for solder joints on the rear of the enclosure.

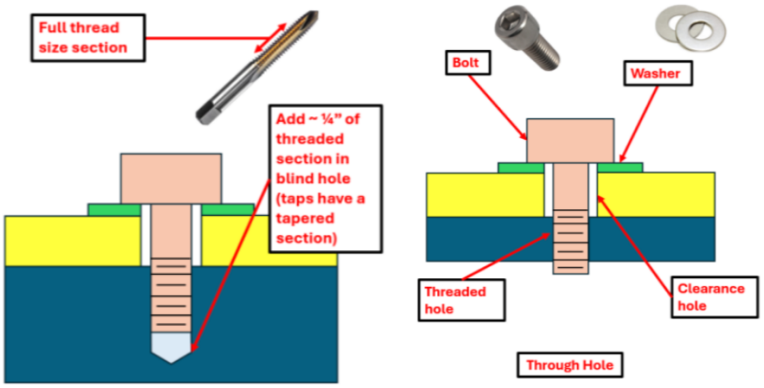


Figure 9: Standards for incorporating fasteners (For more information reference the CAD Procedures document).

Wiring and Space Allocation

I. Provide clearance for wire diameters.
II. Ensure the minimum bend radius of your wires is accounted for.
III. Consider modular wiring for the enclosed component, possibly by including a wire connector.

Mechanical Requirements

I. Minimum wall thickness: 1.6 mm.
II. Shadow lines: 1.6 mm spacing.
III. Increase footprint to accommodate shadow lines without infringing on payload bay constraints.
IV. Allowance for tight-fit battery retention without requiring additional fasteners.

Fastening and Access Strategy

I. Prefer chamfers over fillets to avoid print sag on ceilings.
II. Use counterbore features sized to screw head diameter.
III. Evaluate top-opening vs. side-opening enclosure for accessibility.
IV. Prioritize modularity to access internal components without disassembling other modules.

Heat Inserts and Lugs

I. Reserve adequate material around insert holes (≥1.6 mm).
II. Dimension heat insert diameters and depth according to the Heat Insert Guide.

2.4 Core Onshape Tools Used

I found several Onshape tools very important in developing the enclosure. The sketch tools, particularly construction geometry (keyboard shortcut Q), constraints, and understanding driven vs. driving dimensions, allowed precise control over the design process.

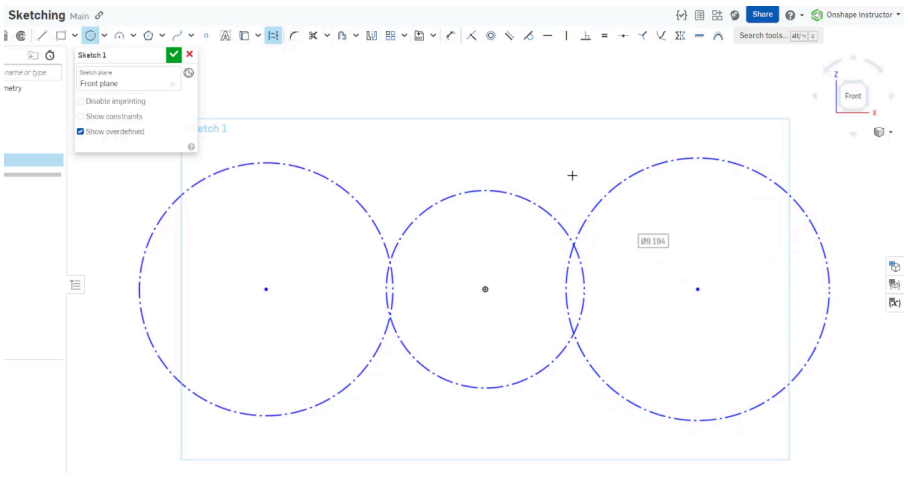


Figure 10: Sketches have a dotted line appearance when using the construction geometry tool.

Understanding constraints such as coincident, tangent, pierce, symmetric, and curvature helped create clean and predictable sketches. I would recommend completing the Onshape Fundamentals course to get comfortable with the interface. Tools like trim, extend, and split were essential when constructing more complex contours or preparing sketches for lofting.

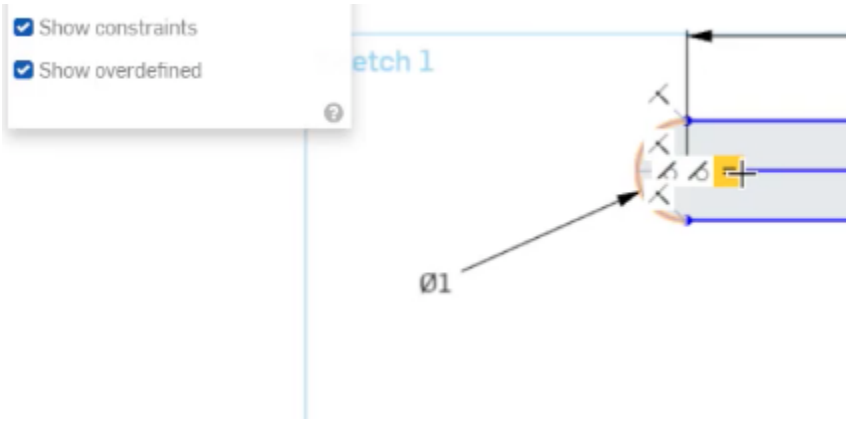


Figure 11: Sketch constraints can be viewed by toggling the checkbox.

The offset tool became especially useful for creating consistent wall thicknesses and shadow lines. Being able to drag-select an entire chain of connected geometry streamlined the process of creating the enclosure. Sketch imprinting and boolean subtract operations helped create internal cavities that maintained alignment with the external geometry. The shell feature could also be used to hollow an enclosure once the perimeter has been fully defined.

For components off the shelf, the McMaster-Carr integration simplifies importing hardware such as screws. After insertion, I fixed these parts in assemblies before using them in derived environments, ensuring they maintained consistent spatial relationships.

Sketching Controls

I. Use Q for construction geometry.
II. Constraints: coincident, tangent, equal, pierce, horizontal/vertical, symmetric, curvature.
III. Toggle between driving (black) and driven (grey) dimensions.
IV. Use trim, extend, and split (especially for loft boundaries).

Offset Tool

I. Offset single loops or multiple connected entities.
II. Use left-click drag to select an entire chain.
III. Useful for projecting and imprinting sketch regions onto faces.

3. Relevance to Career and Team Mission

Developing this Onshape workflow has direct relevance to both my engineering career and our team's mission. As modern engineering increasingly relies on parametric CAD and collaborative cloud based modeling, understanding version controlled master-model workflows is essential. These skills transfer directly to roles in mechanical design, robotics, aerospace hardware, and interdisciplinary engineering teams where multiple contributors need to edit tightly constrained assemblies without breaking downstream dependencies.

For the McMaster Rocketry team, this enclosure is a critical hardware component which directly affects system reliability. Knowing how to design with constraints to avoid unintended rebuild errors improves the robustness and maintainability of our overall payload system.

5. Advice for Future Students Learning This Skill

For anyone learning to build a complex enclosure or any parametric CAD model, the most important lesson I can share is to be intentional about what you base your dimensions and constraints on. Early in this project, I made the mistake of dimensioning directly to the batteries, treating them as fixed geometric references instead of components whose placement might change. When the battery location was updated in Version 3, nearly every sketch and feature tied to those battery edges broke. The cascading errors were so extensive that it was faster to rebuild the entire enclosure from scratch rather than troubleshoot each dependency. It was a time-consuming reminder that references in CAD should be stable, deliberate, and ideally tied to a master sketch or a known condition, not to parts that may shift later in the design.

Along with selecting proper references, I strongly recommend building a master model before creating final enclosure geometry. A nicely structured master sketch with variables gives you a stable backbone that can support design changes without breaking downstream features. When I shifted to this workflow, adjusting clearances or wall thicknesses became a simple parameter change rather than a multi-hour rebuild. Version control is equally essential. When I first referenced the workspace instead of versioned snapshots, unexpected updates propagated into dependent part studios and caused unintended failures. Using versioned references by only updating them when you intend to keeps the design process controlled and predictable.

Finally, resist the urge to add cosmetic features early. Fillets, embossed text, and surface details are often the first features to break and the hardest to repair after updates. Prioritize modeling that locks down structure such as wall thicknesses, mounting hole locations, wiring paths, connector clearances, and internal geometry. Only after these elements are stable should you layer on finishing details.

While the learning curve felt steep, especially after rebuilding the enclosure, it ultimately taught me how to design with more foresight and modularity. These skills make CAD more intuitive and prevent small early decisions from turning into major redesigns down the road.